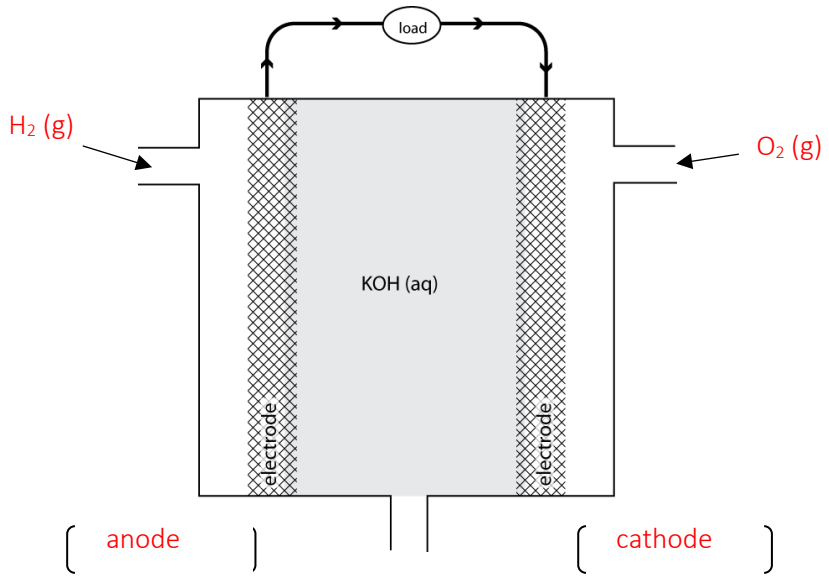
MC questions

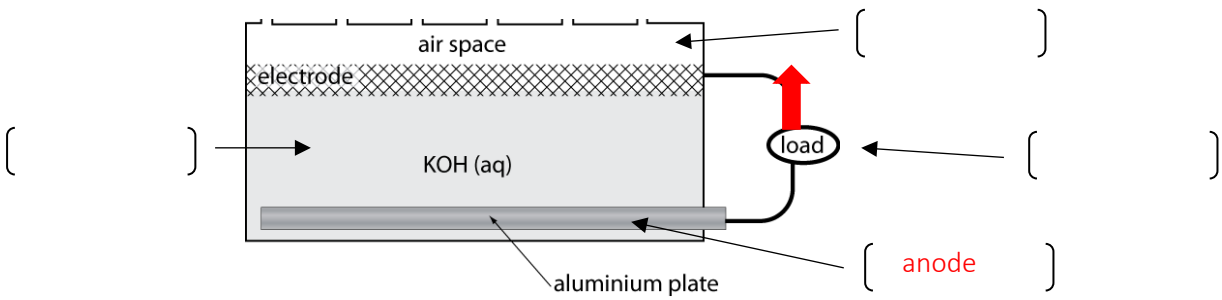
	1	2	3	4	5
V1	D	D	C	B	C
V2	C	A	B	D	B
V3	C	B	D	A	D
V4	B	C	D	B	A
V5	A	D	C	C	B
V6	B	B	D	C	A

6.



(c) the reactants are continually supplied rather than being stored in cell.

7.



(c) $4Al(s) + 3O_2(g) + 6H_2O\ (l) \rightarrow 4Al^{3+}\ (aq) + 12OH^-\ (aq)$ [$4Al(OH)_3$ (aq)/(s) is acceptable]

1 mark for correct reactants 1 mark for correct balancing (consequential if reactants incorrect)

(d) The Co (or Ni) half equation is lower in the ECS and therefore Co (or Ni) is a stronger reducing agent than the oxygen in alkaline solution. 1 mark

In comparison the Ag half reaction is higher in the ECS and therefore a weaker reducing agent than oxygen in alkaline solution. 1 mark

8. (a) Cathode: $Mg^{2+}(l) + 2e^- \rightarrow Mg(l)$ 1 mark

Anode: $2Cl^-(l) \rightarrow Cl_2(g) + 2e^-$ 1 mark Only deduct one mark if states not (l) for one or both.

(b) To prevent molten Mg reacting with oxygen in the air. 1 mark

(c) According to the electrochemical series both Na^+ and Ca^{2+} are weaker oxidants than Mg^{2+} and so are unlikely to interfere with the production of Mg at the cathode. 1 mark

Zn^{2+} is a stronger oxidant than $Mg^{2+}(aq)$ and could be reduced to Zn, thus either preventing the production of Mg or contaminating the Mg produced. 1 mark

(d) According to the electrochemical series Fe is a stronger reductant than Cl^-. At the anode, Fe would be oxidised instead of Cl^- /Fe^{2+} would be produced rather than Cl_2. 1 mark

Half-equation: $Fe(s) \rightarrow Fe^{2+}(l) + 2e^-$ The cations $Fe^{2+}(l)$ would migrate to the cathode/Fe^{2+} is a stronger oxidant than Mg^{2+} hence Fe could be produced/cathode half-equation would be $Fe^{2+}(l) + 2e^- \rightarrow Fe(s)$. 1 mark

9. (a) Rates of forward and reverse reactions are equal ($R_f = R_b$)

The concentrations of reactants and products are constant ([all] constant)

Q = K (any 2 for 2 marks)

(b) (i) Endothermic 1 mark

(ii) Increase in Temp resulted in an increase in product hence K – therefore Endothermic. Or The equilibrium position shifts to partially oppose a change, so when heat was added the solution became bluer (forward reaction) to cause a decrease in temperature (forward reaction consumes energy). Hence the forward reaction is Endothermic. 1 mark

(Consequential mark if incorrect answer in bi is correctly explained in bii)

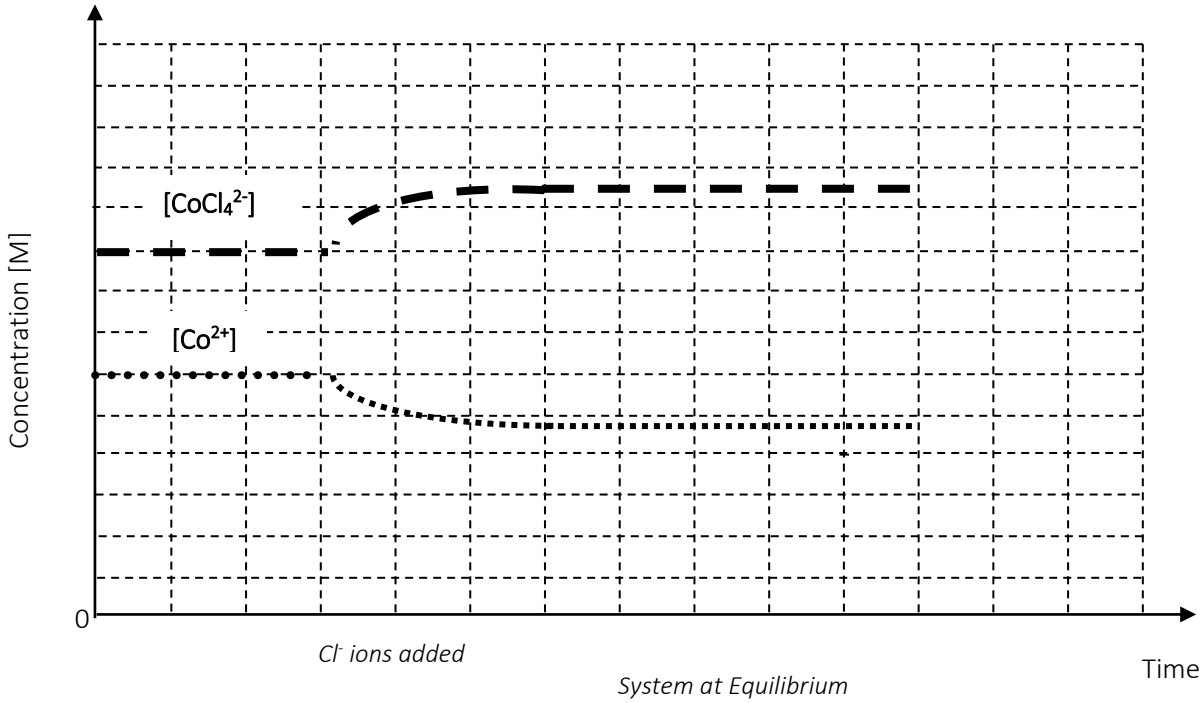
(c) More Pink. 1 mark

(d) The solution became blue because when more Cl^- ions are added they will collide more with the Co^{2+} ions which will initially increase the reaction rate of the forward reaction, causing an equilibrium shift to the right. or use LCP – (shift to partially oppose ie. conc of chloride ion therefore net forward reaction which decreases chloride conc.) or use Q (Q decreases so for reaction to revert to the value of K, Q will increase thereby a shift to the right (blue side of equation) and Q = K.)

(-1/2 if blue colour not mentioned in answer).

(e) no effect as no change in temperature. 1 mark

(f) ½ mark for direction of each curve. 1 mark for both flattening at eqb point. (-1/2 if equilibrium lines not drawn flat)



(g) Net back/ left. (ie. to Co^{2+} (aq) , Cl^-(aq)) 1 mark (Note, "pink" is not a direction…)

(h) If water is added the total concentration of ions in the solution would decrease 1 mark

LCP states that the system would partially oppose this change and produce more particles/ions and increase the total concentration of particles/ ons, therefore net back. 1 mark

NOT decrease in pressure as it is not a gas system.

10.

Version 1	N_2 (g)	$3H_2$ (g)	$2NH_3$ (g)
Initial mol	5.00	6.00	0
Mol reacted	1.60	4.80	3.20 1 mark
Mol equilibrium	3.40	1.20	3.20
[ ] equilibrium	0.85	0.30	0.80

1 mark for correct n(N_2) and n(H_2) 1 mark for correct conversion from n to [ ]

$$K = \frac{[NH_3]^2}{[N_2][H_2]3} = \frac{(0.80)^2}{0.85 \text{ x } (0.30)^3} = 27.9 \ M^{-2}$$

1 mark for correct K (-1 if no or incorrect units)

Version 2	N_2 (g)	$3H_2$ (g)	$2NH_3$ (g)
Initial mol	4.00	5.00	0
Mol reacted	1.10	3.30	2.20 1 mark
Mol equilibrium	2.90	1.70	2.20
[ ] equilibrium	0.967	0.567	0.733

$K = \frac{[NH_3]^2}{[N_2][H_2]3} = \frac{(0.733)^2}{0.967 \times (0.567)^3}$ = 3.06 x 10^{-3} M^{-2} (-1 if no or incorrect units)

Version 3	N_2 (g)	$3H_2$ (g)	$2NH_3$ (g)
Initial mol	4.50	5.50	0
Mol reacted	1.60	4.80	3.20 1 mark
Mol equilibrium	2.90	0.700	3.20
[ ] equilibrium	1.45	0.350	1.60

$K = \frac{[NH_3]^2}{[N_2][H_2]3} = \frac{(1.60)^2}{1.45 \times (0.350)^3}$ = 41.2 M^{-2} (-1 if no or incorrect units)

Version 4	N_2 (g)	$3H_2$ (g)	$2NH_3$ (g)
Initial mol	5.50	6.50	0
Mol reacted	1.55	4.65	3.10 1 mark
Mol equilibrium	3.95	1.85	3.10
[ ] equilibrium	1.32	0.616	1.03

$K = \frac{[NH_3]^2}{[N_2][H_2]3} = \frac{(1.03)^2}{1.32 \times (0.616)^3}$ = 3.43 M^{-2} (-1 if no or incorrect units)

Version 5	N_2 (g)	$3H_2$ (g)	$2NH_3$ (g)
Initial mol	4.25	6.50	0
Mol reacted	1.80	5.40	3.60 1 mark
Mol equilibrium	2.45	1.10	3.60
[ ] equilibrium	1.23	0.55	1.80

$K = \frac{[NH_3]^2}{[N_2][H_2]3} = \frac{(1.80)^2}{1.23 \times (0.55)^3}$ = 15.9 M^{-2}

Version 6	N_2 (g)	$3H_2$ (g)	$2NH_3$ (g)
Initial mol	5.25	6.50	0
Mol reacted	1.45	4.35	2.90 1 mark
Mol equilibrium	3.80	2.15	2.90
[ ] equilibrium	1.27	0.717	0.967

$K = \frac{[NH_3]^2}{[N_2][H_2]3} = \frac{(0.967)^2}{1.27 \times (0.717)^3}$ = 2.00 M^{-2} (-1 if no or incorrect units)

11. For this question, comment on missing or incorrect units, do not deduct marks.

(a) $K = \frac{[CH_3OCH_3]\,[H_2O]}{[CH_3OH]^2}$ 1 mark

	Version 1	Version 2	Version 3	Version 4	Version 5	Version 6
(b)	0.174	0.168	0.173	0.185	0.171	0.168
(c)	0.0170 M	0.0270 M	0.0170 M	0.0185 M	0.0238 M	0.0293 M

(d)	0.0407 M	0.0660 M	0.0407 M	0.0430 M	0.0569 M	0.0714 M
(e)	0.815 mol	1.32 mol	0.815 mol	0.860 mol	1.14 mol	1.43 mol
(f)	1.97 mol	3.18 mol	1.97 mol	2.09 mol	2.75 mol	3.44 mol

Working for version 1:

(b) $K = \frac{1}{5.74}$ = 0.174 (no unit M^0) 1 mark

(c) (i) $[CH_3OCH_3]$ = n/v = 0.340 mol ÷ 20.0 L = 0.0170 M. 1 mark

(-1 if not 3 sig figs)

(ii) $[CH_3OCH_3] = [H_2O] = x$ 1 mark (does not necessarily need to be shown)

$\frac{x^2}{0.0170^2} = 5.74 \rightarrow x^2 = 0.001659 \rightarrow x = 0.0407$ M 1 mark

(iii) $n(CH_3OCH_3)$ = c x v = 0.04073 x 20.0 = 0.815 mol 1 mark

(iv) $n(CH_3OH)$ initial = $n(CH_3OH)$reacted + $n(CH_3OH)$equilibrium

= (0.815 x2 = 1.629) 1 mark

= 1.629 + 0.340 = 1.97 mol 1 mark